Predicting pCR to NAC in Breast Cancer Patients with Logistic Regression

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Abstract

Pathologic Complete Response (pCR) is a critical surrogate endpoint in breast cancer treatment, traditionally used to evaluate the efficacy of neoadjuvant chemotherapies (NAC) prior to surgical intervention. This study utilizes clinical, demographic, and treatment data from 1,506 patients to develop and compare predictive logistic regression models, specifically examining the trade-off between statistical performance and clinical utility. Using Multivariate Imputation by Chained Equations (MICE) to address missingness, we demonstrate that while the inclusion of specific treatment agents in Model 3 provides superior model parsimony and a slight increase in precision, baseline models focused on molecular subtypes (HER2 and Triple Negative) remain the most clinically significant. Model selection based on AIC and Precision-Recall curves indicates that treatment-augmented models refine predictions only after a therapeutic path has been chosen. In contrast, baseline models that exclude treatment variables are of higher clinical importance as they can be utilized at the point of diagnosis to inform the fundamental decision of whether NAC is required at all. By identifying patients with a low probability of pCR regardless of agent, these models provide a framework for prioritizing surgical intervention and avoiding the debilitating side effects of chemotherapy when benefit is statistically minimal.

Keywords: Pathologic Complete Response, Breast Cancer, Logistic Regression, Data Imputation

Introduction

Neoadjuvant chemotherapy (NAC) is standard of care for many breast cancer patients, aiming to shrink tumors prior to surgery. The achievement of pathologic complete response (pCR)—defined as the absence of invasive cancer in the breast and lymph nodes at the time of surgery—is highly correlated with long-term survival. However, response rates vary significantly based on tumor biology and treatment regimen. Predicting which patients will achieve pCR remains a challenge due to the complex interplay between patient demographics, tumor molecular profiles, and varied chemotherapy agents. This project aims to determine the incremental predictive value of adding demographic and treatment-specific information to a baseline clinical model.

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Dataset

This data set contains publicly available clinical and demographic information about breast cancer patients ($N = 1,506$) sourced from four different clinical trials[1]. It includes a robust array of diagnostic indicators essential for assessing treatment efficacy, such as HER2 status and tumor subtype classifications including Triple Negative, Luminal, and HER2-enriched profiles. Clinical assessments also accounted for the presence of multifocal or bilateral disease to capture disease extent. Patient-level demographic variables integrated into the analysis include age, ethnicity, weight, BMI category, and menopausal status. For a significant subset of 1,044 patients, granular NAC data were available, identifying the administration of specific therapeutic agents such as Pembrolizumab, Paclitaxel, Carboplatin, and Trastuzumab.

To maintain the integrity and statistical power of the 1,506-patient sample, Multivariate Imputation by Chained Equations (MICE) ($m = 10$ imputations) was utilized to address missingness, which was most prevalent in the menopause (~33%) and body-mass index (BMI) variables. A Random Forest method was employed to capture non-linear relationships, with predictor variables across including all data except for treatment agents. Those variables being imputed were multifocal cancer status, tumor subtype, HER2 status, BMI group, age, menopausal status, and ethnicity.

Methods

The statistical analysis was conducted using a series of nested logistic regression models designed to evaluate the incremental predictive value of clinical, demographic, and therapeutic variables on the probability of achieving pathologic complete response (pCR). The modeling strategy was divided into three distinct stages:

Model 1: Baseline Clinical Features

The initial model focused exclusively on the biological and structural characteristics of the tumor observed at the time of diagnosis. Variables included BMI group, menopausal status, tumor subtype, HER2 status, multifocal cancer, and bilateral cancer. These were selected as the primary predictors because they define the inherent anatomical features along with the advancement and molecular landscape of the cancer, which traditionally dictate chemotherapy sensitivity.

Model 2: Demographic Integration

The second model expanded upon the baseline by incorporating age and ethnicity. The inclusion of age and ethnicity was critical for adjusting for potential confounding; for instance, the significance of the Luminal subtype only emerged after adjusting for age, suggesting that age-related trends previously masked the independent impact of hormone-receptor-positive biology (see Supplemental Materials for individual variable estimates and significance).

Model 3: Therapeutic Agents

The final model added granularity by including the specific NAC agents prescribed to each patient, such as Pembrolizumab, Carboplatin, and Trastuzumab, pooling agents into a single variable only when they were consistently prescribed together as a treatment regime. These variables were included to determine if the predictive signal of molecular subtypes, particularly Triple Negative (TNBC), was independent of the treatments administered. This model served a mechanistic purpose: to demonstrate how specific high-impact interventions mediate the biological advantages of certain subtypes. While this model has the most predictors, it also constrained the data; only two of the four clinical trials recorded NAC agent prescriptions, meaning a subset ($N^* = 1,044$) of the original population was used for this model.

Logistic Regression Model

The general form of the logistic regression used for each stage is:

$$\text{logit}(P(\text{pCR} = 1)) = \beta^T \mathbf{X}$$

where the coefficients β represent the change in log-odds associated with each predictor, and $\mathbf{X}$ represents the feature encoding for each predictor. Model selection and comparison were performed using Akaike Information Criterion (AIC) and Precision-Recall (PR) curves to evaluate the clinical trade-off between model complexity and predictive utility.

Results

The following sections detail the findings from our nested logistic regression analysis, transitioning from biological baseline predictors to treatment-specific interventions.

Baseline and Demographic Predictors (Models 1 and 2)

Initial analysis of the baseline clinical model revealed that HER2 status, Triple Negative (TNBC) subtype, and BMI underweight were the strongest positive predictors of pCR. Interestingly, BMI demonstrated a non-linear relationship; both underweight and overweight statuses were significant predictors of pCR compared to the normal-weight reference group, whereas the three obesity classes did not reach statistical significance. Upon the inclusion of age and ethnicity in Model 2, the Luminal subtype emerged as a significant negative predictor ($p = 0.03$). This shift indicates that the baseline probability of response in hormone-receptor-positive patients was previously confounded by age-related demographic trends.

Impact of Therapeutic Agents (Model 3)

The inclusion of specific NAC agents in Model 3 fundamentally shifted the model's behavior. Specifically, Triple Negative status lost its statistical significance once high-impact agents such as Pembrolizumab and Carboplatin were added. This "statistical attenuation" indicates that the predictive signal originally attributed to TNBC biology is largely mediated by the specific therapeutic interventions these patients receive. A similar pattern occured for HER2 status, which is used to define tumor subtype in addition to the presence of hormone receptors in the tumor. This again indicates that the predictive power of HER2 status is increasingly captured by the specific anti-HER2 therapies—such as Trastuzumab, Pertuzumab, and T-DM1—included in the expanded model. Similar to the shift seen in TNBC patients, the statistical significance of the biological subtype diminishes when the model accounts for the high-impact therapeutic agents that these patients are specifically targeted to

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receive. Ultimately, this suggests that in contemporary neoadjuvant settings, the achievement of pCR is less a function of inherent tumor biology alone and more a reflection of the specific interaction between that biology and modern targeted NAC agents.

Comparative Results

Here, we highlight the differences and similarities in performance and significance of predictors in each of the three models. As shown in Figure 1, the predictive performance of the models improved incrementally with the addition of more granular data. Model 1 achieved an AUC of 0.70, which increased slightly to 0.71 in Model 2 upon the inclusion of demographic adjustments. Model 3 demonstrated the highest overall performance with an AUC of 0.74, suggesting that while treatment variables provide the most precise fit, the baseline clinical and demographic features capture the majority of the predictive signal for pathologic complete response. It is important to note that in Figure 1, the ROC curves are generated from the largest data set applicable to each model; Models 1 and 2 were fit on the full population ($N = 1,506$) while Model 3 was fit on the subset ($N^* = 1,044$).

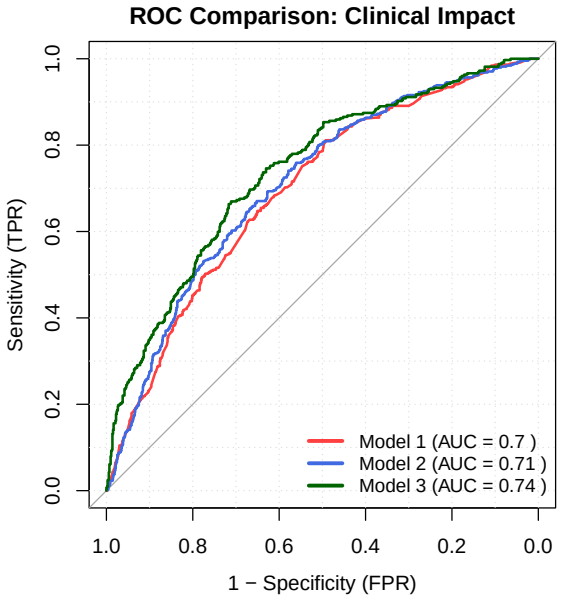

Figure 1: ROC Curves Comparison

The comparative forest plot in Figure 2 reveals that clinical and treatment factors are the primary drivers of pathologic complete response, with underweight status notably emerging as one of the top three strongest predictors. While luminal subtype persists as the only significant negative predictor across all models, the superior performance of Model 3 is underscored by the fact that neoadjuvant agents constitute seven of the top nine significant positive predictors across models (with seven out the top eight for Model 3 in particular), whereas ethnicity remains non-significant throughout the analysis.

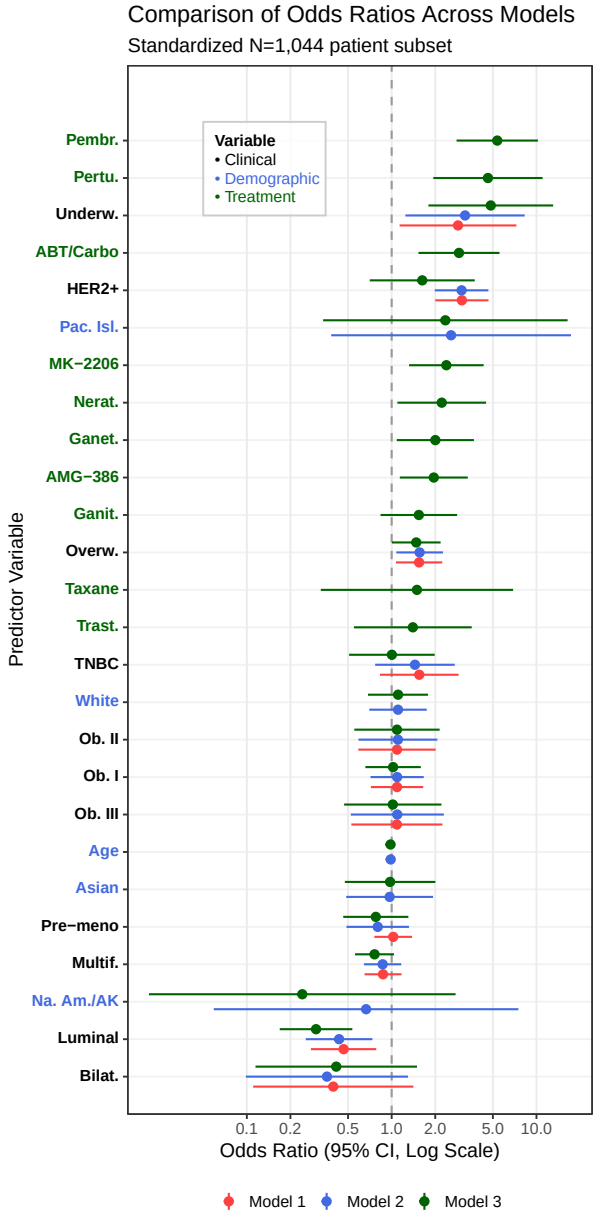

Figure 2: Odds-Ratio Comparison

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Model Selection and Predictive Utility

To evaluate the clinical trade-offs between these models, we compared their performance using AIC and PR curves on the standardized patient subset ($N^* = 1,044$). As show in Table 1, Model 3 demonstrated the lowest mean AIC, confirming it as the most parsimonious model for explaining the variance in pCR outcomes. Furthermore, the PR curve analysis in Figure 3 showed that the treatment-augmented model provided a distinct "lift" in precision, maintaining higher positive predictive values across various recall thresholds compared to the clinical-only models. However, as the baseline models maintained a competitive Area Under the ROC Curve (AUROC), they remain robust tools for initial screening.

Table 1: Akaike Information Criterion Comparison

Model	Description	Mean.AIC
Model 1	Baseline	1220.7
Model 2	Demographic	1223.9
Model 3	**Treatment**	**1192.6**

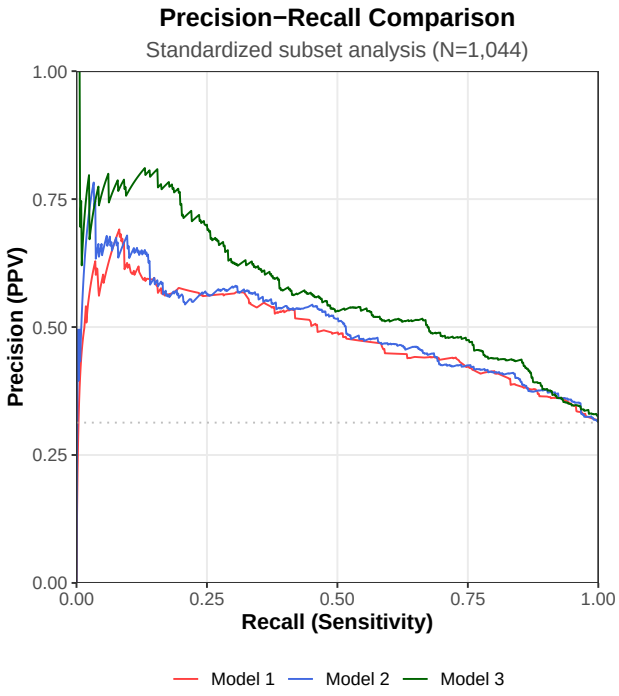

Figure 3: Precision Recall Curve Comparison

Clinical Significance of the Baseline Model

Despite the minor performance advantages of Model 3, the results underscore the vital importance of the baseline and demographic models. While the treatment-inclusive model provides a refined retrospective view of response, the baseline model offers the highest utility for pre-treatment decision-making. By identifying patients with a low baseline probability of pCR before any agents are administered, clinicians can more accurately weigh the risks of chemotherapy toxicity against the likelihood of successful response.

Conclusion

The comparative analysis of these three models reveals a critical distinction between statistical optimization and clinical utility. While the inclusion of neoadjuvant treatment agents in Model 3 provides the highest predictive precision and the most parsimonious fit according to AIC metrics, this performance gain must be weighed against practical application in a pre-treatment setting. The treatment-augmented model is no predictive, but inherently retrospective; it confirms that specific drugs drive the pathologic response, effectively mediating the biological signals of HER2 status and subtypes like Triple Negative.

However, for the practicing clinician, the baseline and demographic models (Models 1 and 2) hold greater clinical importance. By relying solely on information available before a treatment plan is finalized, these models provide the necessary foresight to evaluate whether a patient is likely to benefit from neoadjuvant chemotherapy at all. Our results indicate that while accounting for patient age reveals a significant negative association between Luminal breast cancer status and pCR, the minute increase in predictive power and worsened AIC score indicate that the baseline model is a more parsimonious choice.

In cases where the baseline model predicts a negligible probability of pCR, clinicians may consider bypassing systemic therapy in favor of immediate surgical intervention. This approach prioritizes patient quality of life by avoiding the significant toxicities and long-term side effects of chemotherapy when the likelihood of a complete pathologic response is statistically minimal. Ultimately, while the inclusion of specific treatment agents increases the model's predictive "ceiling," the true value lies in identifying who can safely undergo less intensive therapy, thereby reducing the medical burdens on breast cancer patients.

Gemini 3 was used to generate code for some plots and correct grammar.

References

1. Garrucho L, Kushibar K, Reidel C-A, Joshi S, Osuala R, Tsirikoglou A, et al. A large-scale multicenter breast cancer DCE-MRI benchmark dataset with expert segmentations. Scientific Data 2025;12:453. https://doi.org/10.1038/s41597-025-04707-4.

Supplemental Materials

Table 2: Model 1 - Pooled Logistic Regression

Variable	estimate	std.error	statistic	df	P_Value
(Intercept)	**-1.574**	**0.275**	**-5.719**	**1335.758**	**0.000**
multifocal_cancer	-0.109	0.132	-0.832	476.424	0.406
bilateral_breast_cancer	-1.026	0.557	-1.842	1490.747	0.066
bmi_groupobesity_class_1	0.011	0.180	0.061	857.202	0.951
bmi_groupobesity_class_2	0.129	0.263	0.490	290.248	0.625
bmi_groupobesity_class_3	-0.117	0.292	-0.401	551.110	0.689
bmi_groupoverweight	**0.461**	**0.154**	**2.992**	**648.354**	**0.003**
bmi_groupunderweight	**1.233**	**0.376**	**3.275**	**362.539**	**0.001**
her2	**1.237**	**0.206**	**6.000**	**1454.648**	**0.000**
tumor_subtypeluminal	-0.405	0.215	-1.880	1432.443	0.060
tumor_subtypetriple_negative	**0.837**	**0.264**	**3.163**	**1324.640**	**0.002**
menopausepre	0.121	0.129	0.937	590.632	0.349

Table 3: Model 2 - Pooled Logistic Regression (with Demographics)

Variable	estimate	std.error	statistic	df	P_Value
(Intercept)	-1.089	0.601	-1.813	617.600	0.070
multifocal_cancer	-0.113	0.133	-0.850	444.089	0.396
bilateral_breast_cancer	-1.086	0.560	-1.937	1484.602	0.053
bmi_groupobesity_class_1	0.056	0.184	0.303	1004.906	0.762
bmi_groupobesity_class_2	0.198	0.270	0.733	268.015	0.464
bmi_groupobesity_class_3	-0.030	0.301	-0.100	545.358	0.920
bmi_groupoverweight	**0.492**	**0.156**	**3.160**	**662.387**	**0.002**
bmi_groupunderweight	**1.294**	**0.378**	**3.425**	**396.908**	**0.001**
age	-0.012	0.009	-1.334	373.728	0.183
her2	**1.218**	**0.208**	**5.861**	**1431.585**	**0.000**
tumor_subtypeluminal	**-0.475**	**0.219**	**-2.171**	**1374.021**	**0.030**
tumor_subtypetriple_negative	**0.782**	**0.268**	**2.921**	**1240.794**	**0.004**
menopausepre	-0.060	0.204	-0.295	189.513	0.768
ethnicityamerican indian/alaska native	-0.588	1.105	-0.532	1440.171	0.595
ethnicityasian	0.222	0.305	0.729	1339.217	0.466
ethnicitycaucasian	0.298	0.179	1.660	1111.091	0.097
ethnicityhawaiian/pacific islander	**1.980**	**0.811**	**2.440**	**1466.871**	**0.015**
ethnicityhispanic	-0.921	0.861	-1.070	866.769	0.285

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Table 4: Model 3 - Pooled Logistic Regression (Clinical + Treatment Agents)

Variable	estimate	std.error	statistic	df	P_Value
(Intercept)	-14.365	1005.594	-0.014	1010.006	0.989
bilateral_breast_cancer	-0.881	0.654	-1.346	1005.164	0.179
multifocal_cancer	-0.272	0.158	-1.719	994.444	0.086
her2	0.487	0.425	1.144	911.573	0.253
tumor_subtypeluminal	**-1.202**	**0.294**	**-4.088**	**976.305**	**0.000**
tumor_subtypetriple_negative	0.004	0.347	0.010	958.755	0.992
age	-0.018	0.012	-1.528	251.025	0.128
bmi_groupobesity_class_1	0.023	0.224	0.101	544.808	0.919
bmi_groupobesity_class_2	0.085	0.344	0.247	172.273	0.806
bmi_groupobesity_class_3	0.018	0.394	0.045	553.430	0.964
bmi_groupoverweight	**0.391**	**0.197**	**1.988**	**279.465**	**0.048**
bmi_groupunderweight	**1.576**	**0.504**	**3.127**	**299.760**	**0.002**
menopausepre	-0.252	0.262	-0.962	147.460	0.337
ethnicityamerican indian/alaska native	-1.421	1.242	-1.144	1008.889	0.253
ethnicityasian	-0.025	0.367	-0.067	964.967	0.946
ethnicitycaucasian	0.102	0.243	0.418	882.084	0.676
ethnicityhawaiian/pacific islander	0.854	0.991	0.862	987.896	0.389
ethnicityOther/Unknown	-14.722	756.652	-0.019	1010.006	0.984
agent_paclitaxel	14.168	1005.594	0.014	1010.005	0.989
agent_ganitumab	0.432	0.310	1.394	1006.380	0.163
agent_ganetespib	**0.695**	**0.312**	**2.224**	**1004.136**	**0.026**
agent_mk_2206	**0.870**	**0.302**	**2.877**	**993.059**	**0.004**
agent_pembrolizumab	**1.679**	**0.330**	**5.096**	**998.047**	**0.000**
agent_amg_386	**0.670**	**0.275**	**2.436**	**998.777**	**0.015**
agent_trastuzumab	0.337	0.477	0.707	830.615	0.480
agent_neratinib	**0.797**	**0.359**	**2.217**	**942.059**	**0.027**
agent_t_dm1	14.420	1005.594	0.014	1010.006	0.989
agent_pertuzumab	**1.531**	**0.443**	**3.459**	**1000.229**	**0.001**
agent_anthracycline	13.861	1005.594	0.014	1010.006	0.989
agent_taxane	0.402	0.779	0.516	839.804	0.606
*agent_taxane___incomplete_*	-15.610	860.703	-0.018	1010.006	0.986
agent_abt_888_carboplatin	**1.071**	**0.328**	**3.264**	**1001.530**	**0.001**

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Code Appendix

```r
knitr::opts_chunk$set(echo = F)

library(tidyverse)
library(dplyr)
library(recipes)
library(kableExtra)
library(gtsummary)
library(ggplot2)
library(pROC)
library(precrec)
library(mice)
library(ggtext)

clinicalData <- read.csv("clinical_and_imaging_info.csv") %>%
  mutate(
    pcr = replace_na(pcr, replace=0)
  )
relevantData <- clinicalData %>%
  select(
    pcr, patient_id,
    dataset, nac_agent,
    bilateral_breast_cancer,
    multifocal_cancer,
    hr, er, pr, her2,
    nottingham_grade,
    tumor_subtype,
    age, weight, bmi_group,
    menopause, ethnicity
  )

# clean missing data
relevantData <- relevantData %>%
  mutate(across(where(is.character), ~na_if(str_trim(.), "")))

# clean up data
relevantData <- relevantData %>% select(
  patient_id, pcr, dataset, nac_agent,
  bilateral_breast_cancer,
  multifocal_cancer,
  her2, tumor_subtype,
  age, bmi_group,
  menopause, ethnicity
) %>% mutate(
  menopause = case_when(
    str_detect(menopause, "(?i)pre|peri") ~ "pre",
    str_detect(menopause, "(?i)post")     ~ "post",
    is.na(menopause)                      ~ NA_character_,
  ),
  tumor_subtype = case_when(
    str_detect(tumor_subtype, "(?i)luminal") ~ "luminal",
    str_detect(tumor_subtype, "(?i)her2")    ~ "her2_enriched",
    is.na(tumor_subtype)                     ~ NA_character_,
```

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```r
    TRUE                                        ~ tumor_subtype
  ),

  ethnicity = case_when(
    str_detect(ethnicity, "(?i)american indian|native") ~ "american indian/alaska native",
    str_detect(ethnicity, "(?i)hawai|pacific islander") ~ "hawaiian/pacific islander",
    str_detect(ethnicity, "(?i)multiple race")          ~ NA_character_,
    is.na(ethnicity)                                    ~ NA_character_,
    TRUE                                                ~ ethnicity
  )
)

# Now one-hot encode treatment variables
unique_agents <- relevantData %>%
  filter(!is.na(nac_agent) & nac_agent != "") %>%
  pull(nac_agent) %>%
  str_split(" \\+ ") %>%  # Splits by " + "
  unlist() %>%
  str_trim() %>%
  unique()

for (agent in unique_agents) {
  # Create a clean column name
  col_name <- paste0("agent_", str_replace_all(tolower(agent), "[^a-z0-9]", "_"))

  relevantData <- relevantData %>%
    mutate(!!col_name := case_when(
      is.na(nac_agent) ~ NA_real_,
      stringr::str_detect(as.character(nac_agent), stringr::fixed(agent)) ~ 1,
      TRUE ~ 0
    ))
}

relevantData <- relevantData %>% select(-nac_agent)

write.csv(relevantData, "./processedData.csv", row.names=F)

data <- read.csv("./processedData.csv")
dataNoMissing <- data %>% drop_na()

# Impute the missing data with Multiple Imputations by Chained Equations (MICE)
data_for_imputation <- data %>%
  mutate(across(where(is.character), as.factor)) %>%
  select(-starts_with("agent_"))

# We need to remove patient id as a predictor for MICE
ini <- mice(data_for_imputation, maxit = 0)

# Setting a column to 0 in the predictorMatrix means it won't be used to predict anything else
pred <- ini$predictorMatrix
pred[, "patient_id"] <- 0  # patient_id will not be used as a predictor
pred["patient_id", ] <- 0  # patient_id itself will not be imputed

# Setting the method to "" tells mice to ignore this column during the math
```

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```r
meth <- ini$method
meth["patient_id"] <- ""

imputed_data <- mice(data_for_imputation,
                     m = 10,
                     method = meth,
                     predictorMatrix = pred,
                     seed = 123)

# Run your logistic regression on all 5 datasets and pool the results
fit1 <- with(imputed_data,
             glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                 bmi_group + her2 + tumor_subtype + menopause,
                 family = binomial))
pooled_results1 <- pool(fit1)

# Run your logistic regression on all 5 datasets and pool the results
fit2 <- with(imputed_data,
             glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                  bmi_group + age + her2 + tumor_subtype +
                  menopause + ethnicity, family = binomial))
pooled_results2 <- pool(fit2)

# Add NAC agent columns to the prediction, but don't include in imputation
agents_to_join <- data %>%
  mutate(.id = as.integer(row_number())) %>%
  select(.id, starts_with("agent_"))

subset_long_data <- complete(imputed_data, action = "long", include = TRUE) %>%
  left_join(agents_to_join, by = ".id") %>%
  filter(dataset %in% c("ISPY2", "NACT")) %>%
  mutate(ethnicity = as.character(ethnicity),
         # Group Hispanic into 'Other' since it is sparse in this subset
         ethnicity = ifelse(ethnicity == "hispanic", "Other/Unknown", ethnicity),
         ethnicity = as.factor(ethnicity)) %>%
  select(pcr, .imp, .id, everything()) %>%
  mutate(.imp = as.integer(.imp),
         .id = as.integer(.id))

agent_cols <- names(subset_long_data)[grep("agent_", names(subset_long_data))]

# Now we combine collinear agents so our model doesn't break

# Calculate Correlation Matrix
cor_matrix <- cor(subset_long_data[, agent_cols], use = "pairwise.complete.obs")

processed <- c()
collapsed_data <- subset_long_data

for (col in agent_cols) {
  if (col %in% processed) next

  # A. Identify Positive Collinearity (r > 0.99)
  identical_cols <- names(which(cor_matrix[col, ] > 0.99))
```

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```r
  # B. Identify Anti-Collinearity (r < -0.99)
  anti_cols <- names(which(cor_matrix[col, ] < -0.99))

  if (length(identical_cols) > 1) {
    # Combine Positive Collinearity
    new_name <- paste(identical_cols[1],
                      paste(gsub("agent_", "", identical_cols[-1]), collapse = "_"),
                      sep = "_")
    collapsed_data <- collapsed_data %>%
      mutate(!!new_name := get(col)) %>%
      select(-all_of(identical_cols))
    processed <- c(processed, identical_cols)

  } else if (length(anti_cols) > 0) {
    # Combine Anti-Collinearity
    new_name <- paste0(col, "_vs_", gsub("agent_", "", anti_cols[1]))

    collapsed_data <- collapsed_data %>%
      mutate(!!new_name := get(col)) %>%
      select(-all_of(c(col, anti_cols)))

    processed <- c(processed, col, anti_cols)
  }
}

# Now we can put the new columns into the pre-imputed datasets
imputed_collapsed <- as.mids(collapsed_data)
new_agent_vars <- names(collapsed_data)[grep("agent_", names(collapsed_data))]
final_agent_vars <- new_agent_vars[!new_agent_vars %in% c("agent_fec100")]
formula_str <- paste("pcr ~ bilateral_breast_cancer + multifocal_cancer + her2 +
                     tumor_subtype + age + bmi_group + menopause + ethnicity +",
                     paste(final_agent_vars, collapse = " + "))

# Finally, fit the model
fit3 <- with(imputed_collapsed, glm(as.formula(formula_str), family = binomial))
pooled_results3 <- pool(fit3)


predictions <- list()
for(i in 1:imputed_data$m) {
  # Extract the i-th completed dataset
  dat <- complete(imputed_data, i)

  # Fit the model on that specific dataset
  m_fit <- glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                 bmi_group + her2 + tumor_subtype + menopause,
               data = dat, family = binomial)

  predictions[[i]] <- predict(m_fit, type = "response")
}

avg_probs_m1 <- rowMeans(as.data.frame(predictions))
```

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```r
actual_pcr <- complete(imputed_data, 1)$pcr
roc_curve1 <- roc(actual_pcr, avg_probs_m1)
m1_auc <- pROC::auc(roc_curve1)


predictions <- list()
for(i in 1:imputed_data$m) {
  # Extract the i-th completed dataset
  dat <- complete(imputed_data, i)

  # Fit the model on that specific dataset
  m_fit <- glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                  bmi_group + age + her2 + tumor_subtype +
                  menopause + ethnicity,
               data = dat, family = binomial)

  predictions[[i]] <- predict(m_fit, type = "response")
}

avg_probs_m2 <- rowMeans(as.data.frame(predictions))

roc_curve2 <- roc(actual_pcr, avg_probs_m2)
m2_auc <- pROC::auc(roc_curve2)


predictions <- list()
for(i in 1:imputed_collapsed$m) {
  # Extract the i-th completed dataset (which already has agents and filtering)
  dat <- complete(imputed_collapsed, i)

  # Fit the treatment model (Model 3)
  m_fit <- glm(pcr ~ bilateral_breast_cancer + multifocal_cancer +
                 her2 + tumor_subtype + age + bmi_group +
                 menopause + ethnicity +
                 agent_paclitaxel + agent_ganitumab + agent_ganetespib +
                 agent_mk_2206 + agent_pembrolizumab + agent_amg_386 +
                 agent_trastuzumab + agent_neratinib + agent_abt_888_carboplatin +
                 agent_pertuzumab + agent_taxane,
               data = dat, family = binomial)
  predictions[[i]] <- predict(m_fit, type = "response")
}

avg_probs_m3 <- rowMeans(as.data.frame(predictions))
actual_pcr_subset <- complete(imputed_collapsed, 1)$pcr
roc_curve3 <- roc(actual_pcr_subset, avg_probs_m3)
m3_auc <- pROC::auc(roc_curve3)

par(pty = "s") # Square plot area
plot(roc_curve1,
     col = "brown1",
     lwd = 2,
     main = "ROC Comparison: Clinical Impact",
     xlab = "1 - Specificity (FPR)",
     ylab = "Sensitivity (TPR)",
```

Gemini 3 was used to generate code for some plots and correct grammar.

```r
     grid = TRUE)

# 2. Add the subsequent curves to the same plot
lines(roc_curve2, col = "royalblue", lwd = 2)
lines(roc_curve3, col = "darkgreen", lwd = 2)

# 3. Add a legend to distinguish the models and show AUCs
legend("bottomright",
       legend = c(paste("Model 1 (AUC =", round(m1_auc, 2), ")"),
                  paste("Model 2 (AUC =", round(m2_auc, 2), ")"),
                  paste("Model 3 (AUC =", round(m3_auc, 2), ")")),
       col = c("brown1", "royalblue", "darkgreen"),
       lwd = 3,
       bty = "n",       # No border around legend
       cex = 0.9)       # Slightly smaller font for the legend

# Refit models 1 and 2 on the subset of data to get fair AIC comparisons
target_patient_ids <- complete(imputed_collapsed, 1)$patient_id

fit1_refit <- with(imputed_data,
                   glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                       bmi_group + her2 + tumor_subtype + menopause,
                       subset = patient_id %in% target_patient_ids,
                       family = binomial))

fit2_refit <- with(imputed_data,
                   glm(pcr ~ multifocal_cancer + bilateral_breast_cancer +
                       bmi_group + her2 + tumor_subtype + menopause + age +
                       ethnicity,
                       subset = patient_id %in% target_patient_ids,
                       family = binomial))
# Plot OR forest plot with all three models together
# Helper function to extract and transform pooled results
get_plot_data <- function(fit, model_name) {
  df <- as.data.frame(summary(pool(fit), conf.int = TRUE))
  df <- df[df$term != "(Intercept)", ]
  data.frame(
    term = df$term,
    OR = exp(df$estimate),
    low = exp(df$`2.5 %`),
    high = exp(df$`97.5 %`),
    Model = model_name
  )
}

# Combine all three models
plot_data_combined <- rbind(
  get_plot_data(fit1_refit, "Model 1"),
  get_plot_data(fit2_refit, "Model 2"),
  get_plot_data(fit3, "Model 3")
)

# remove variables with high sparsity/extreme estimates
plot_data_filtered <- plot_data_combined[
```

Gemini 3 was used to generate code for some plots and correct grammar.

```r
  plot_data_combined$high < 50 & plot_data_combined$low > 0.02,
]

# rename variables to make the plot fit better
plot_data_filtered <- plot_data_filtered %>%
  mutate(term = recode(term,
    "multifocal_cancer" = "Multif.",
    "bilateral_breast_cancer" = "Bilat.",
    "bmi_groupobesity_class_1" = "Ob. I",
    "bmi_groupobesity_class_2" = "Ob. II",
    "bmi_groupobesity_class_3" = "Ob. III",
    "bmi_groupoverweight" = "Overw.",
    "bmi_groupunderweight" = "Underw.",
    "her2" = "HER2+",
    "tumor_subtypeluminal" = "Luminal",
    "tumor_subtypetriple_negative" = "TNBC",
    "menopausepre" = "Pre-meno",
    "age" = "Age",
    "ethnicityamerican indian/alaska native" = "Na. Am./AK",
    "ethnicityasian" = "Asian",
    "ethnicitycaucasian" = "White",
    "ethnicityhawaiian/pacific islander" = "Pac. Isl.",
    "ethnicityhispanic" = "Hisp.",
    "ethnicityOther/Unknown" = "Eth. Unk.",
    "agent_paclitaxel" = "Pacli.",
    "agent_ganitumab" = "Ganit.",
    "agent_ganetespib" = "Ganet.",
    "agent_mk_2206" = "MK-2206",
    "agent_pembrolizumab" = "Pembr.",
    "agent_amg_386" = "AMG-386",
    "agent_trastuzumab" = "Trast.",
    "agent_neratinib" = "Nerat.",
    "agent_t_dm1" = "T-DM1",
    "agent_pertuzumab" = "Pertu.",
    "agent_anthracycline" = "Anthr.",
    "agent_taxane" = "Taxane",
    "agent_taxane__incomplete_" = "Taxane (Inc)",
    "agent_abt_888_carboplatin" = "ABT/Carbo"
  ))

plot_data_filtered <- plot_data_filtered %>%
  mutate(Type = case_when(
    term %in% c("Multif.", "Bilat.", "HER2+", "Luminal", "TNBC", "Pre-meno",
                "Ob. I", "Ob. II", "Ob. III", "Overw.", "Underw.") ~ "Clinical",
    term %in% c("Age", "White", "Asian", "Hisp.",
                "Pac. Isl.", "Na. Am./AK", "Eth. Unk.") ~ "Demographic",
    TRUE ~ "Treatment"
  )) %>%
  mutate(color_term = case_when(
    Type == "Clinical" ~ paste0("<span style='color:black;'>", term, "</span>"),
    Type == "Demographic" ~ paste0("<span style='color:royalblue;'>", term, "</span>"),
    Type == "Treatment" ~ paste0("<span style='color:darkgreen;'>", term, "</span>")
  ))
```

Gemini 3 was used to generate code for some plots and correct grammar.

```r
# Plot it
ggplot(plot_data_filtered, aes(x = reorder(color_term, OR), y = OR, color = Model)) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "grey60", linewidth = 0.8) +
  geom_pointrange(aes(ymin = low, ymax = high),
                  position = position_dodge(width = 0.8),
                  size = 0.5,
                  linewidth = 0.8) +
  coord_flip() +
  theme_bw(base_size = 16) +
  scale_x_discrete(expand = expansion(add = c(1, 2))) +
  scale_y_log10(breaks = c(0.1, 0.2, 0.5, 1, 2, 5, 10)) +
  scale_color_manual(values = c("brown1", "royalblue", "darkgreen")) +

  # Manual "Legend" for Y-axis colors in the top left
  annotate("rect", xmin = 24, xmax = 26.5, ymin = 0.05, ymax = 0.35,
           fill = "white", color = "grey80", alpha = 0.8) +
  annotate("text", x = 26, y = 0.06, label = "Variable",
           fontface = "bold", hjust = 0, size = 4.5) +
  annotate("text", x = 25.5, y = 0.06, label = "• Clinical",
           color = "black", hjust = 0, size = 4.5) +
  annotate("text", x = 25, y = 0.06, label = "• Demographic",
           color = "royalblue", hjust = 0, size = 4.5) +
  annotate("text", x = 24.5, y = 0.06, label = "• Treatment",
           color = "darkgreen", hjust = 0, size = 4.5) +

  theme(
    legend.position = "bottom",
    legend.title = element_blank(),
    axis.text.y = element_markdown(face = "bold"),
    panel.grid.major.y = element_line(color = "grey95"),
    panel.grid.minor = element_blank(),
    plot.margin = margin(10, 20, 10, 10)
  ) +
  labs(
    title = "Comparison of Odds Ratios Across Models",
    subtitle = "Standardized N=1,044 patient subset",
    x = "Predictor Variable",
    y = "Odds Ratio (95% CI, Log Scale)"
  )

# Tabulate AIC comparison
aic_table_data <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3"),
  Description = c("Baseline", "Demographic", "Treatment"),
  `Mean AIC` = c(
    mean(sapply(fit1_refit$analyses, AIC)),
    mean(sapply(fit2_refit$analyses, AIC)),
    mean(sapply(fit3$analyses, AIC))
  )
)

# Render table
kable(aic_table_data,
      digits = 1,
```

Gemini 3 was used to generate code for some plots and correct grammar.

```r
      booktabs = TRUE,
      caption = "Akaike Information Criterion Comparison") %>%
  kable_styling(latex_options = c("striped", "hold_position")) %>%
  row_spec(3, bold = TRUE, background = "#e5f5e0")

# Create Precision-Recall Curve
get_avg_probs <- function(mira_fit) {
  m_count <- length(mira_fit$analyses)
  pred_list <- lapply(mira_fit$analyses, function(x) predict(x, type = "response"))
  rowMeans(as.data.frame(pred_list))
}

avg_probs_m1 <- get_avg_probs(fit1_refit)
avg_probs_m2 <- get_avg_probs(fit2_refit)

precrec_obj <- evalmod(
  scores = list(avg_probs_m1, avg_probs_m2, avg_probs_m3),
  labels = list(actual_pcr_subset, actual_pcr_subset, actual_pcr_subset),
  modnames = c("Model 1", "Model 2", "Model 3")
)

# Plot the Comparison
autoplot(precrec_obj, curvetype = "PRC", size = 1.2) +
  theme_bw() +
  coord_cartesian(expand = FALSE) +
  theme(
    legend.position = "bottom",
    legend.title = element_blank(),
    text = element_text(size = 12),
    plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5, color = "grey30"),
    panel.grid.minor = element_blank(),
    axis.title = element_text(face = "bold"),
    plot.margin = unit(c(0.2, 0.4, 0.1, 0.2), "cm")
  ) +
  scale_color_manual(values = c("brown1", "royalblue", "darkgreen")) +
  labs(
    title = "Precision-Recall Comparison",
    subtitle = "Standardized subset analysis (N=1,044)",
    x = "Recall (Sensitivity)",
    y = "Precision (PPV)"
  )

reg_table_data <- summary(pooled_results1) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(Variable = term, P_Value = p.value)

# 2. Identify rows where p-value < 0.05
sig_rows <- which(reg_table_data$P_Value < 0.05)

# 3. Create the table
reg_table_data %>%
  kbl(caption = "Model 1 - Pooled Logistic Regression",
      booktabs = TRUE) %>%
```

Gemini 3 was used to generate code for some plots and correct grammar.

```r
  kable_styling(bootstrap_options = c("striped", "hover"),
                full_width = FALSE,
                latex_options = "HOLD_position") %>%
  row_spec(sig_rows, bold = TRUE) %>% # Bolds significant predictors
  column_spec(1, italic = TRUE)

reg_table_data <- summary(pooled_results2) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(Variable = term, P_Value = p.value)

# 2. Identify rows where p-value < 0.05
sig_rows <- which(reg_table_data$P_Value < 0.05)

# 3. Create the table
reg_table_data %>%
  kbl(caption = "Model 2 - Pooled Logistic Regression (with Demographics)",
      booktabs = TRUE) %>%
  kable_styling(bootstrap_options = c("striped", "hover"),
                full_width = FALSE,
                latex_options = "HOLD_position") %>%
  row_spec(sig_rows, bold = TRUE) %>% # Bolds significant predictors
  column_spec(1, italic = TRUE)

reg_table_data <- summary(pooled_results3) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(Variable = term, P_Value = p.value)

# 2. Identify rows where p-value < 0.05
sig_rows <- which(reg_table_data$P_Value < 0.05)

# 3. Create the table
reg_table_data %>%
  kbl(caption = "Model 3 - Pooled Logistic Regression (Clinical + Treatment Agents)",
      booktabs = TRUE) %>%
  kable_styling(bootstrap_options = c("striped", "hover"),
                full_width = FALSE,
                latex_options = "HOLD_position") %>%
  row_spec(sig_rows, bold = TRUE) %>% # Bolds significant predictors
  column_spec(1, italic = TRUE)
```

Gemini 3 was used to generate code for some plots and correct grammar.